
Integrable Systems — Exercise Sheet 1

Phase space and flows

Variant: Questions only

Conventions, as in Lecture 1: on $T^*\mathbb{R}^n$ the Hamiltonian vector field is $X_H = \sum_i (\partial_{p_i} H \partial_{q_i} - \partial_{q_i} H \partial_{p_i})$; the flow of a field X is φ_t^X ; the Lie bracket is $[X, Y]f = X(Yf) - Y(Xf)$, with components $[X, Y]^i = X(Y^i) - Y(X^i)$.

Exercise 1 (charts and phase spaces). (i) Write points of S^1 as $(\cos a, \sin a)$. Let U be S^1 without its top point $(0, 1)$, with the angle coordinate $\theta_U \in (-3\pi/2, \pi/2)$, and V be S^1 without its bottom point $(0, -1)$, with $\theta_V \in (-\pi/2, 3\pi/2)$. Show that $U \cap V$ consists of two arcs, and that $\theta_V - \theta_U$ is constant on each, with different constants. (ii) Give the configuration space, the phase space and its dimension for the spherical pendulum (a bob on a rigid massless rod, free to swing in every direction about a fixed point), the double pendulum (in a plane), and a free rigid body. (iii) A classical spin of fixed length has as its states the directions in space, the sphere S^2 . Take as given: S^2 is compact; the cotangent bundle T^*Q of a manifold Q of dimension $n \geq 1$ is the set of pairs (a point q of Q , a momentum at q), the map $T^*Q \rightarrow Q$ forgetting the momentum is continuous, and the momenta at one point q form a closed subset of T^*Q that is a copy of $\mathbb{R}^n$ (Lectures 2 and 4 build this properly); a closed subset of a compact space is compact; and continuous maps take compact sets to compact sets. Show that S^2 is not a cotangent bundle T^*Q of any manifold Q of positive dimension.

Exercise 2 (flows, and completeness). (i) On $T^*\mathbb{R}$ take $H = qp$. Find the flow of X_H , check that $\varphi_t \circ \varphi_s = \varphi_{t+s}$, and say whether X_H is complete. (ii) Now take $H = q^2 p$. Solve Hamilton's equations from (q_0, p_0) , and find for which t the solution exists. Check that H is constant along it. (iii) Let X be a vector field on a manifold that conserves a smooth function F ($XF = 0$) all of whose level sets are compact. Show that X is complete. (iv) Show that the only smooth functions conserved by $\dot{x} = x^2$ on $\mathbb{R}$ are the constants, and that the conserved H of (ii) has non-compact level sets. Show that, by contrast, the pendulum $H = \frac{1}{2}p^2 - \cos \theta$ has compact level sets on its cylinder. What does (iii) conclude in each case?

Exercise 3 (a Lie bracket, and two flows that do not commute). On $\mathbb{R}^2$ with coordinates (q, p) let $X = \partial_q$ and $Y = q \partial_p$. (i) Compute $[X, Y]$. Show that $X = X_H$ for $H = p$ and $Y = X_K$ for $K = -\frac{1}{2}q^2$. (ii) Find the two flows, and compute $\varphi_s^Y \circ \varphi_t^X$ and $\varphi_t^X \circ \varphi_s^Y$. Are they equal? Is this consistent with the lecture's theorem on commuting flows? (iii) Compute $\varphi_{-s}^Y \circ \varphi_{-t}^X \circ \varphi_s^Y \circ \varphi_t^X(q, p)$: follow X , then Y , then X backwards, then Y backwards. Compare the result with $ts [X, Y]$.

Exercise 4 (three phase portraits). For $H = \frac{1}{2}p^2 + V(q)$ on $T^*\mathbb{R}$, with

$$(a) V = \frac{1}{4}q^4 - \frac{1}{2}q^2, \quad (b) V = \frac{1}{3}q^3 - q, \quad (c) V = -1/\cosh^2 q,$$

find the equilibria and their energies. For each range of energies E , say how many orbits the level set $H = E$ consists of and which are closed curves; name the levels that contain a separatrix; and say whether any orbit leaves every bounded set in finite time.

Exercise 5 (the period of the rotating pendulum). For the pendulum $H = \frac{1}{2}p^2 - \cos \theta$ on its cylinder and $E > 1$, the lecture says the rotating orbit with $p > 0$ is periodic. Let $K(k) = \int_0^{\pi/2} d\varphi / \sqrt{1 - k^2 \sin^2 \varphi}$ for $0 \leq k < 1$. (i) Show that the period is $T(E) = \int_0^{2\pi} d\theta / \sqrt{2(E + \cos \theta)}$, and that $T(E) = 2k K(k)$ with $k^2 = 2/(E + 1)$. (ii) Show that $2\pi/\sqrt{2(E + 1)} < T(E) < 2\pi/\sqrt{2(E - 1)}$, and that $T(E)\sqrt{2E} \rightarrow 2\pi$ as $E \rightarrow \infty$. What motion is that limit? (iii) Show that $T(E) \rightarrow \infty$ as $E \rightarrow 1^+$. (iv) Evaluate $T(2)$ numerically.

Exercise 6 (Liouville's theorem on phase-space volume). Let X be a vector field on $\mathbb{R}^m$ with flow φ_t , and $J(t, x) = \det D\varphi_t(x)$, the Jacobian determinant of $x \mapsto \varphi_t(x)$. (i) Show that $\partial_t D\varphi_t(x) = DX(\varphi_t(x)) D\varphi_t(x)$, and deduce $\partial_t J = (\operatorname{div} X)(\varphi_t(x)) J$, so that $J(t, x) = \exp \int_0^t (\operatorname{div} X)(\varphi_s(x)) ds$. (ii) Show that $\operatorname{div} X_H = 0$ on $T^*\mathbb{R}^n$ for every smooth H , and deduce *Liouville's theorem*: the flow of X_H preserves phase-space volume, $\operatorname{vol}(\varphi_t(R)) = \operatorname{vol}(R)$, for every region R and every time t for which φ_t is defined on all of R . (iii) For $H = q^2 p$ (Exercise 2), at which points is φ_1 defined? Why does the theorem need its qualification? (iv) For $H = qp$, find the image of the unit disc under φ_t . What is preserved, and what is not? (v) For the damped oscillator $\dot{x} = v$, $\dot{v} = -x - \gamma v$ with $\gamma > 0$, which is not Hamiltonian, find J .

Exercise 7 ★. *The bracket is the failure of two flows to commute.* Let X and Y be smooth vector fields on $\mathbb{R}^m$, with flows φ_t^X and φ_s^Y . (i) Show that $\varphi_t^X(x) = x + tX(x) + \frac{1}{2}t^2 DX(x)X(x) + O(t^3)$. (ii) Show that

$$\varphi_{-s}^Y \circ \varphi_{-t}^X \circ \varphi_s^Y \circ \varphi_t^X(x) = x + ts [X, Y](x) + O((|t| + |s|)^3).$$

(iii) Check (ii) against Exercise 3 (iii). (iv) Deduce one half of the lecture's theorem on commuting flows: if $\varphi_t^X \circ \varphi_s^Y = \varphi_s^Y \circ \varphi_t^X$ for all small s, t , then $[X, Y] = 0$.